

Real-Time Marginal Ambiguity Covariance for Tightly Coupled GNSS–Visual–Inertial Factor Graphs

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Abstract—Carrier-phase ambiguity resolution (AR) in tightly coupled GNSS/IMU/vision estimators consumes a float ambiguity covariance Q_{aa} . Open platforms often substitute a GNSS-only “shadow” covariance because forming the joint sliding-window covariance online is considered too expensive, discarding the IMU and visual information that tight coupling is meant to provide. We present an exact-in-window marginal $Q_{aa} = [H_{\text{active}}^{-1}]_{aa}$ obtained by two-stage sparse Schur elimination over the active GNSS–IMU–vision window, including the marginalization prior, with a numerical self-validation gate and shadow fallback. On 1 787 open-sky AR epochs the method produces a validated covariance on 99.94% of epochs, with 90.1% achieving relative Frobenius error $\|Q_{\text{prop}} - Q_{\text{Ceres}}\|_F / \|Q_{\text{Ceres}}\|_F$ below 10^{-3} ; among the remaining disagreements a trace proxy is larger (conservative) in 171 of 177 cases and within 7.3×10^{-4} in 6 near-ties. Mean covariance query time is 28 ms versus 812 ms for Ceres ($\sim 29\times$), with median 27 ms, p95 49 ms, p99 59 ms, and 4.3% of epochs above a 50 ms covariance budget (max 97.5 ms). End-to-end effects versus shadow AR are reported as secondary system impact; an overconfident local-conditioning approximation is harmful. Full matrix Loewner diagnostics and a Ceres end-to-end oracle are required to support a stronger “exact” claim and are in progress (Sec. IX).

Index Terms—RTK, ambiguity resolution, factor graph, marginal covariance, GNSS/IMU/vision, Schur complement

I. Introduction

Tightly coupled GNSS/IMU/vision estimators resolve carrier-phase integers using a float ambiguity covariance fed to LAMBDA-type search and acceptance tests. In GICI-LIB [1], the open RTK/IMU/camera reference, AR consumes a covariance from a parallel GNSS-only shadow estimator because a full-graph query via `ceres::Covariance` is too slow online (mean 812 ms on our filled-window open-sky graphs; 99.6% of epochs exceed 50 ms). The shadow path discards IMU and visual information at the moment AR needs it most.

Cheaper alternatives that condition on neighboring states without forming the Schur complement are fast but can be overconfident: on UrbanNav Deep, a local-conditioning variant raised fix rate while degrading horizontal RMSE from 2.77 ± 0.44 m to 6.75 ± 2.98 m. AR therefore needs a covariance that is both consistent with the joint window and affordable every epoch.

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This paper focuses on that covariance problem. We make three contributions:

- 1) An algorithm that extracts the exact-in-window marginal ambiguity covariance $Q_{aa} = [H_{\text{active}}^{-1}]_{aa}$ by structure-exploiting two-stage elimination, without inverting the full information matrix and without dropping the marginalization prior.
- 2) A numerical self-validation gate with typed abstention and shadow fallback, so an unreliable factorization never feeds AR.
- 3) A covariance-level evaluation against shadow AR, local conditioning, and `ceres::Covariance`, with end-to-end system impact reported secondarily.

The two evaluation questions are: (i) does the proposed path reproduce the Ceres ambiguity covariance (and, offline, Ceres-driven AR decisions)? (ii) by how much does it reduce cost relative to the estimator/AR cycle? Trajectory RMSE is secondary and is not the headline claim.

II. Related Work

a) Integer estimation.: ILS via LAMBDA [2] / MLAMBDA [3], ratio tests and FFRT [4], [5], partial AR, and bootstrapping success rates P_s [6] are standard. BIE [7], [8] softens hard commits. Our focus is the float covariance that those methods consume inside a multi-sensor graph, not a new integer estimator.

b) GNSS/IMU/vision fusion.: Tight VIO [9], [10] and GNSS–VIO systems [1], [11] optimize sliding-window factor graphs with marginalization priors. GICI-LIB’s AR path uses a GNSS-only shadow covariance—the gap we close at the covariance interface.

c) Covariance recovery in factor graphs.: Marginal covariances are classical in Gauss–Newton estimation. Sparse factorization and selected inversion recover subsets of H^{-1} without forming the full dense inverse [12], [13]. Incremental smoothers (iSAM2 / Bayes tree) maintain square-root information and can extract covariances from the clique structure [14], [15]. Ceres exposes sparse-QR covariance queries [16] that are exact for the linearized problem but costly on large live windows. Our method is a domain-structured elimination for the ambiguity block of a GNSS/IMU/vision RRR window: landmark Schur first, then dense reduction of the remaining nuisance states, with an AR-oriented abstention policy.

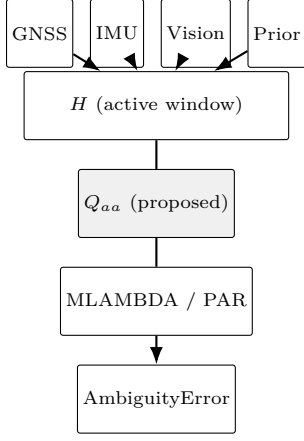


Fig. 1. RRR window and AR path. The shaded block is the proposed exact-in-window marginal; the baseline uses a GNSS-only shadow covariance instead.

d) Urban GNSS.: NLOS/multipath mitigations include masks, robust losses, shadow matching [17], and 3DMA [18], [19]. UrbanNav-HK [20] provides Medium/Deep benchmarks. We use these datasets to stress end-to-end AR under the proposed Q_{aa} ; we do not claim a new NLOS classifier.

III. Problem Formulation

We consider a Ceres-based sliding-window RRR graph with double-differenced GNSS residuals, IMU preintegration, reprojection factors, and a dense marginalization prior. Let x collect active states and $H = \sum_r J_r^\top J_r$ the Gauss-Newton information in minimal coordinates (Triggs-corrected Jacobians for robust residuals), including the prior information Λ . Partition $x = (a, \ell, n)$ into ambiguities a , landmarks ℓ , and remaining nuisance n (poses, speed/bias, clocks, frequencies, extrinsics, prior-only variables).

AR requires the marginal

$$Q_{aa} = [H^{-1}]_{aa}. \quad (1)$$

ceres::Covariance computes (1) at whole-problem sparse-QR cost. The shadow estimator replaces (1) with a GNSS-only approximation. A local-conditioning shortcut drops the Schur term involving out-of-set neighbors and is fast but can understate uncertainty.

a) Requirements.: (R1) Exactness in the active window: Q_{aa} must equal the Ceres marginal up to linearization and numerics. (R2) Real-time: cost bounded by window size, with a fallback when factorization is unreliable. (R3) AR safety preference: when numerics disagree, prefer larger uncertainty over silent overconfidence.

Fig. 1 places Q_{aa} in the AR path.

IV. Proposed Marginal Covariance Method

a) Assembly.: We form H from all active residuals plus the marginalization prior through a read-only accessor exposing $\Lambda = J^\top J$ with per-block minimal offsets (generic residual evaluation of the prior is avoided under live window growth).

Algorithm 1 Exact-in-window marginal ambiguity covariance

Require: active residuals, prior Λ , ambiguity index set a

- 1: Assemble $H \leftarrow \sum_r J_r^\top J_r$ (Triggs) $+\Lambda$
- 2: for each landmark block ℓ do
- 3: Rank-truncated Schur-eliminate ℓ from H
- 4: end for
- 5: Jacobi-equilibrate nuisance; LDLT factorize
- 6: Iterative refine; if gate fails then return shadow Q
- 7: return $Q_{aa} \leftarrow [H_{\text{red}}^{-1}]_{aa}$

b) Two-stage elimination.: No residual couples an ambiguity to a landmark ($H_{a\ell} = 0$; prior-touched landmarks are promoted into n), so

$$H = \begin{bmatrix} H_{aa} & H_{an} & 0 \\ H_{na} & H_{nn} & H_{n\ell} \\ 0 & H_{\ell n} & H_{\ell\ell} \end{bmatrix}. \quad (2)$$

Stage 1 forms the landmark Schur update

$$\bar{H}_{nn} = H_{nn} - H_{n\ell} H_{\ell\ell}^+ H_{\ell n} \quad (3)$$

via independent 3×3 rank-truncated pseudo-inverses. Exactness condition: null modes of each H_{ii} have zero coupling rows in H_{*i} , so truncating those modes does not change the (a, n) marginal. Stage 2 then yields the ambiguity marginal

$$Q_{aa} = \left(H_{aa} - H_{an} \bar{H}_{nn}^{-1} H_{na} \right)^{-1} \quad (4)$$

by Jacobi-equilibrated dense LDLT on $\bar{H}_{nn}$ plus iterative refinement (Fig. 2, Alg. 1).

c) Relation to standard covariance recovery.: Ceres selected covariance [16] asks for the same blocks but factors the whole problem (sparse QR) each query. Sparse selected inversion / Takahashi methods recover entries of H^{-1} from a global factorization [12], [13]; Bayes-tree methods reuse incremental square-root information [14]. Our contribution is not a new linear-algebra primitive but a domain-structured online path: landmark blocks are eliminated at $O(1)$ each, the remaining GNSS/IMU/extrinsic/prior nuisance is small enough for dense reduction every AR epoch, the marginalization prior is included through a read-only Λ accessor, and a typed abstention prevents feeding AR an untrusted Q_{aa} . Complexity: $O(L)$ for L landmarks (3×3 each) plus $O(n_n^3 + n_a n_n^2)$ for nuisance size n_n and ambiguity size n_a , bounded by the sliding window.

d) Self-validation and fallback.: After LDLT, a few refinement steps on the reduced system are accepted only if the last increment's effect on the Schur complement is below the scale of the smallest eigenvalue of the reduced nuisance block (no dataset-tuned thresholds). Abstention cases: non-finite assembly, failed factorization, failed gate, or non-PD reduced ambiguity information. On abstention the estimator returns the shadow covariance for that epoch.

The estimator is registered as an opt-in variant (rtk_imu_camera_rrr_va); upstream defaults are

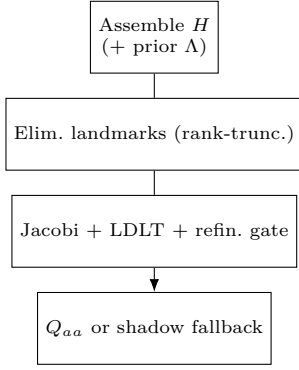


Fig. 2. Two-stage elimination with abstention (F2).

TABLE I
Covariance vs. ceres::Covariance (GICI-board 1.1)

Metric	Value
Validated output (fast_ok)	99.94% of 1787 (1 abstention)
Frobenius rel. err. $\leq 10^{-3}$	90.1% of usable
Frobenius disagreements $> 10^{-3}$	177
trace larger (proxy)	171
trace below-ref. (near-tie)	6 (worst 7.3×10^{-4})
ASan (1761 epochs)	clean

unchanged. File-mode UrbanNav configs disclose `max_age=35s` (for 30s HKKT base epochs) and `relative_frequency=0.01` (code default); neither was tuned against UrbanNav ground truth.

V. Numerical Validation and Complexity

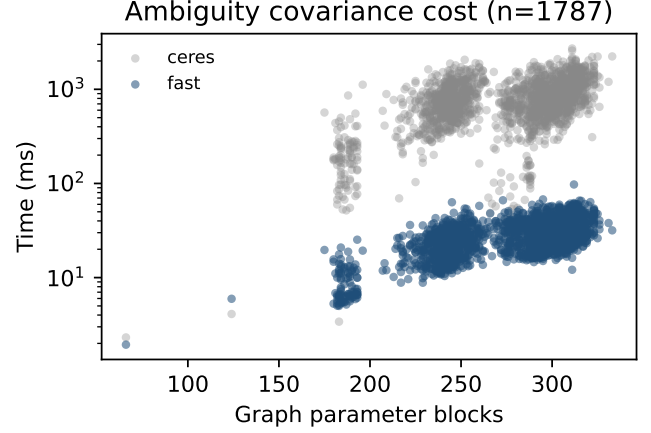
a) Reference.: Per-AR-epoch comparison against `ceres::Covariance` on identical live graphs (GICI-board 1.1, 1787 epochs), requesting the same ambiguity blocks (not a whole-state inverse straw man). The primary match metric already logged online is the normalized Frobenius error $\|Q_{\text{prop}} - Q_{\text{Ceres}}\|_F / \|Q_{\text{Ceres}}\|_F$. Trace comparisons are a secondary safety proxy only. Loewner / directional diagnostics ($\text{mineig}(Q_{\text{prop}} - Q_{\text{Ceres}})$, generalized eigenvalues of the pair) are instrumented in the benchmark path and must be reported before a stronger exactness claim (Sec. IX).

b) Accuracy.: Table I: a validated covariance is produced on 99.94% of epochs (one abstention)—this is not a 99.94% match rate. Of usable epochs, 90.1% have Frobenius relative error $\leq 10^{-3}$. Of 177 larger Frobenius disagreements, a trace proxy is larger in 171 cases and within 7.3×10^{-4} in 6 near-ties. Larger trace does not imply Loewner dominance in every direction.

c) Runtime.: Table II and Fig. 3 report the covariance query only. AR in this stack is invoked near GNSS epoch rate ($\sim 1\text{Hz}$); a 50ms covariance budget is a soft online target ($\sim 5\%$ of a 1Hz slot), not a hard 20Hz estimator deadline. Mean proposed query 28ms ($\sim 29\times$ vs Ceres); 4.3% exceed 50ms (max 97.5ms). Total optimize+covariance latency will be reported with confirmatory runs.

TABLE II
Runtime on the same 1787 AR epochs (ms)

Method	mean	med	p95	p99	max	> 50ms
Proposed	28.0	26.7	49.1	58.8	97.5	4.3%
Ceres	811.9	758.1	1573	1892	2714	99.6%

Fig. 3. Proposed vs. Ceres wall time versus graph size (GICI-board 1.1, $n = 1787$).TABLE III
Datasets

Dataset	Role	DOY	Notes
GICI-board 1.1	Cov. + runtime	—	Open-sky
UrbanNav Deep	System impact	141 Whampoa	$n=15119$
UrbanNav Med.	Float-path (held)	iso. 137 TST	pending isolation

d) Complexity.: Stage 1 is linear in the number of landmarks with 3×3 work each. Stage 2 is cubic in the dense nuisance dimension after landmark elimination, hence bounded by the sliding window rather than trajectory length—the same structural reason the method tracks Ceres’ marginal while remaining cheaper than a whole-problem sparse QR each epoch.

VI. Experimental Setup

UrbanNav Deep uses the session-aligned 5s HKKT base slice `hkkt141g.21o` (not the full-day 30s file). Primary metric: unaligned raw ENU RMSE (E/N/U, yaw). Sim(3) APE is used only when comparing to Chi Table V. Admissibility: Deep $n_{\text{matched}} \geq 14000$; Medium ≥ 6500 . Research flags are opt-in and default-off; the frozen upstream baseline APE on GICI-board 1.1 ($0.029198\text{m} / 0.470557^\circ$, bands $\pm 0.005\text{m} / \pm 0.1^\circ$) is re-checked after builds.

Post-file replay is load-sensitive (same Deep sequence spans $\sim 14\text{--}24\text{min}$ wall under identical configs). We treat this as a reproducibility limitation (Sec. IX), not a scientific contribution; confirmatory UrbanNav numbers below use identical-load pairing or solo-idle execution.

TABLE IV
Deep horizontal ablation (raw ENU; one protocol per row)

Covariance for AR	Deep h (m)	Fix rate
Shadow (paired w/ local study)	2.77 ± 0.44	$\sim 1.3\%$
Local conditioning (unsafe)	6.75 ± 2.98	$\sim 17\%$
Shadow (paired w/ proposed)	3.03 ± 0.68	1.7%
Proposed exact-in-window	3.40 ± 0.78	$8.8 \pm 1.2\%$
Ceres exact (oracle e2e)	—	pending

Spreads are repeat-run variation on one Deep sequence, not independent scenes. Ceres oracle end-to-end is required before claiming proposed reproduces Ceres AR.

TABLE V
Deep system impact: proposed Q_{aa} vs. shadow (one sequence, 3 repeats)

Arm	h (m)	u (m)	yaw ($^\circ$)	fix
Shadow	3.03 ± 0.68	2.27 ± 0.27	1.09 ± 0.28	1.7%
Proposed	3.40 ± 0.78	1.46 ± 0.21	0.66 ± 0.07	8.8%

VII. Covariance Accuracy and Runtime

Open-sky end-to-end with the proposed path preserves accuracy and fix count inside the frozen band: $0.028991\text{ m} / 0.4727^\circ$, 869/1775 fixed versus baseline 876/1775. Thus, when the measurement model is well behaved, joint Q_{aa} neither inflates fixing nor loses fixes relative to shadow AR.

The local-conditioning approximation is the critical negative control (Table IV): it is real-time but unsafe for AR on Deep. The proposed path removes that catastrophic horizontal regression while keeping a large fraction of the fix-rate increase that joint information enables.

VIII. End-to-End AR Evaluation

a) System impact of the covariance swap.: Table V reports paired Deep runs that differ in the AR covariance path (proposed vs. shadow), under the same protocol wave. Vertical RMSE and yaw improve in all three repeats; fix rate rises about fivefold; horizontal RMSE is mixed (worse in 2/3 repeats). These repeats are not independent statistical samples; they measure run-to-run variation under load-sensitive replay. Medium is withheld from result tables until float-path isolation passes (Sec. IX): with 0% NIFix, continuous trajectories must match before Deep deltas can be attributed solely to AR.

b) What still needs an oracle.: The decisive end-to-end claim is that proposed AR decisions and trajectories match a Ceres-fed AR oracle at much lower covariance cost. That offline oracle is configured but not yet reported here. Fig. 4 shows only that swapping to joint Q_{aa} can change fix timing and later horizontal error; it is not a causal proof of prior poisoning.

IX. Limitations

- Exactness evidence incomplete. Frobenius match at 10^{-3} covers 90.1% of usable epochs; Loewner PSD checks, per-diagonal errors, and worst-case directional

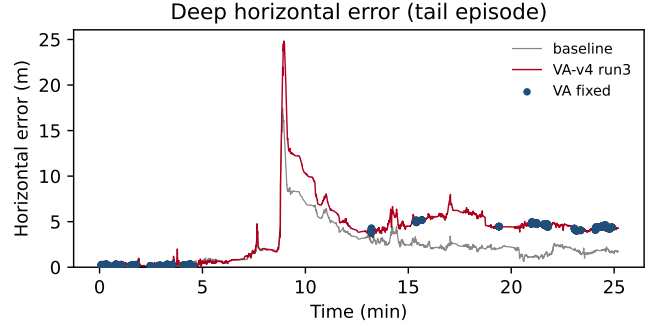


Fig. 4. Example Deep horizontal error over time (one proposed-path run with a heavy tail vs. a shadow baseline run). Markers: fixed epochs on the proposed arm.

variance ratios are instrumented and must be reported before claiming elementwise equivalence to Ceres.

- No Ceres e2e oracle yet. Covariance-level agreement does not imply identical AR decisions on UrbanNav; an offline Ceres-fed AR run is required.
- Float-path isolation pending. Arm-to-arm Medium differences at 0% NIFix indicate solver-timing / WIFix / ordering confounders until a deterministic isolation test passes.
- Soft real-time. The 50 ms figure is a soft covariance budget at $\sim 1\text{ Hz}$ AR; 4.3% exceed it (max 97.5 ms). Report optimize+cov latency.
- Statistics. Three repeats of one Deep sequence are not three independent environments.
- Decision-layer urban mitigations are outside this letter.

X. Conclusion

We presented a domain-structured online path to the exact-in-window marginal ambiguity covariance of a tightly coupled GNSS-IMU-vision factor graph, with numerical abstention and shadow fallback. On open-sky graphs it is about $29\times$ faster than requesting the same blocks from `ceres::Covariance`, produces a validated output on 99.94% of epochs, and achieves Frobenius relative error below 10^{-3} on 90.1% of usable epochs. Stronger Loewner diagnostics, a Ceres end-to-end AR oracle, and float-path isolation are the remaining measurements needed before submission-strength exactness and causality claims. Secondary Deep system effects versus shadow AR (more fixes; mixed horizontal) must be interpreted in that light.

Reproducibility

Opt-in flags, configs, and evaluation scripts are in the accompanying repository (`research/paper/README.md`). Runtime percentiles regenerate from `[vaar-fast]` logs via `scripts/fig_timing.py`. Table generation: `scripts/eval_paper_tables.py`.

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